A FLAT FAMILY OVER A DISCRETE VALUATION RING WITH REDUCED SPECIAL FIBRE HAS CONNECTED GENERIC FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8131-8146 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8148-8163, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Flat morphisms over a discrete valuation ring, connectedness, and reducedness of the special fibre; the argument is retained.

Source reference. Original lemma 0538: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8163–8163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A flat degeneration of a disconnected scheme is either disconnected or nonreduced. Let $f : X \to S$ be a morphism of schemes. Assume that*

(1) *S is the spectrum of a discrete valuation ring,*
(2) *f is flat,*
(3) *X is connected,*
(4) *the closed fibre X_s is reduced.*

Then the generic fibre X_η is connected.

Proof. Write $S = \mathrm{Spec}(R)$ and let $\pi \in R$ be a uniformizer. To get a contradiction assume that X_η is disconnected. This means there exists a nontrivial idempotent $e \in \Gamma(X_\eta, \mathcal{O}_{X_\eta})$. Let $U = \mathrm{Spec}(A)$ be any affine open in X. Note that π is a nonzerodivisor on A as A is flat over R, see More on Algebra, Lemma 0538 for example. Then $e|_{U_\eta}$ corresponds to an element $e \in A[1/\pi]$. Let $z \in A$ be an element such that $e = z/\pi^n$ with $n \geq 0$ minimal. Note that $z^2 = \pi^n z$. This means that $z \bmod \pi A$ is nilpotent if $n > 0$. By assumption $A/\pi A$ is reduced, and hence minimality of n implies $n = 0$. Thus we conclude that $e \in A$! In other words $e \in \Gamma(X, \mathcal{O}_X)$. As X is connected it follows that e is a trivial idempotent which is a contradiction. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.